In-Order Pipelines

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In-order.

- In-order Fetch ✓
- In-order Decode ✓
- In-order execution ✓
- In-order completion ?

In-order Commit

IF ID EX M W
IF ID E M
IF ID E

FP insn. vs. integer insn

DIV

⇓

Diversified pipelines

1. Unpipelined FP Units

IF ID EX MEM WB

FADD → 3 cycles

Ex: FADD f1, f2, f3
FADD f4, f5, f6

IF ID EX1 EX2 EX3 M
IF ID ID/s ID/s EX1
IF IF/s IF/s ID
IF

structural hazard

FADD f1 f2 f3 IF ID EX1 EX2 E3 M W

```
FADD  f1, f2, f3      IF  ID  EX1  EX2  E3   M   W
ADD   R1, R2, R3          IF  ID   EX   M   W
ADD   R4, R5, R6              IF   ID   EX  S   M
ADD   R11, R12, R13               IF   ID  S
                                   IF  S
```

2. Pipeline FP Units

Structural hazards ↕

WAW data hazard

WAR

RAW.

- MUL/ADD ← R1

MUL/ADD R1 ←

Ex: FPADD: 3 cycle latency ADD 0 cycle - bypass, OOO completion

```
nop  f1 = f2 + f3      IF | ID | EX1 | EX2 | E3 | E4 | M  | WB
nop  f4 = f1 + f5          | IF | ID  | (S) | S  | S  | E1 | E2 | E3 | E4 | M | WB
nop                        |    |     |     |    |    | ID | ?
```

idp
n.p

f4 = f1 + f5

r1 = r2 + r3

IF	ID	(S)	S	S	E1	E2	E3	E4	M	WB
	IF	(S)	S	S	ID	?				
						E	M	WB		

E M WB (in completion)

pipeline stalls ⟶ correctness

└→ Who is responsible for generating stalls? < HW / SW → nops

Variable - Latency Instruction

ld r1, #4(r0) // r1 = mem[r0+4]

↓ virtual

ld r1, #4(r0) (\$ miss)

ld r2, #8(r4) (\$ hit)

add r5, r1, #4

add r10, r11, r12

. stall on miss

IF	ID	(EX)	Mem	S	S	S	S	W	
	IF	ID	EX	M	WB				
		IF	ID	(S)				E	
			IF	S				ID	
								IF	

\# miss ↑

stall-on-use

S	...	WB	
S		M	
S		E	
S		ID	
		IF	

- stall on use
  - OOO completion
  - MSHR (miss status handing register)
    - 1 MSHR tracks one cache line access